E-Track: Eye Tracking with Event Camera for Extended Reality (XR) Applications

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***Abstract*—Eye tracking is an essential functionality to enable extended reality (XR) applications. However, the latency and power constraints of an XR headset are tight. Unlike fix-rate frame-based RGB cameras, the event camera senses brightness changes and generates asynchronous sparse events with high temporal resolution. Although the event camera exhibits suitable characteristics for eye tracking in XR systems, processing an event-based data stream is a challenging task. In this paper, we present an event-based eye-tracking system that extracts pupil features. It is the first system that operates only with an event camera and requires no additional sensing hardware. We first propose an event-to-frame conversion method that encodes the events triggered by eye motion into a 3-channel frame. Secondly, we train a Convolutional Neural Network (CNN) on 24 subjects to classify the events representing the pupil. Finally, we employ a region of interest (RoI) mechanism that tracks pupil location and reduces the amount of CNN inference by 96%. Our eye-tracking pipeline is able to locate the pupil with an error of 3.68 pixels at 160 mW system power.**

***Index Terms*—Eye Tracking, Event Camera, Neuromorphic Hardware, Convolutional Neural Network, Extended Reality**

I. Introduction

XR headsets are the next-generation mobile platform with applications spanning from education [1] to healthcare [2]. To build a lightweight system that provides a real-time immersive experience all day, improvement in performance and power efficiency is required. One of the crucial functional blocks in an XR system is eye tracking which enables gaze estimation for foveated rendering [3], varifocal display [4], and gaze UI [5]. On an XR headset, a near-eye camera captures the visual information of one eye in the field of view (FoV), and eye tracking locates and tracks the eye and its components.

Although RGB cameras are commonly used in current systems [6]–[8], they are not ideal for eye tracking due to their fixed-rate operation. As the fastest moving organ in the human body, the location of the pupil can change drastically between contiguous frames, thereby causing the system to lose track of the eye. On the other hand, while the eye is not moving, the recorded frame serves no purpose. Additionally, RGB frame is a dense data format, which is disadvantageous for a system with low power and low latency requirements. An emerging alternative is the event camera: a retina-inspired neuromorphic sensor. Events are generated only when the brightness change at a pixel location exceeds a threshold. In contrast to RGB camera, event camera has the advantage of being movement-triggered, having a high temporal resolution, and having a sparse data format that better suits the eye-tracking task.

However, the existing image processing and computer vision methods developed for frame-based data are not directly applicable to event-based data. The processing of event data format is an active research topic [9]. The event-based eye tracking system proposed by [10] utilizes both RGB camera and event camera for frame and event data stream. The system requires the pixels classified in the frame and the events that are close to eye parts to perform tracking. The system in [11] requires multiple IR LEDs to generate glints in the FoV to track the eye. Both these approaches require additional hardware setup that increases the headset size and power consumption and can not operate with only the event camera.

We propose an eye-tracking system to address the above issue. To the best of our knowledge, it is the first fully event-based eye-tracking system that requires no additional hardware. Specifically, we make the following contributions:

- We develop an event-to-frame conversion method to encode event features into a 3-channel frame.
- We train a low latency segmentation CNN model on 24 subjects to identify the events that represent the pupil.
- We design an event-based RoI mechanism to track the pupil that operates asynchronously and reduces system workload.

II. Background

A. Event Camera

Event camera [12], also known as neuromorphic camera, is a dynamic vision sensor. The pixels in the event camera are constantly sensing the brightness changes independently and asynchronously. When the intensity difference at a pixel location exceeds a threshold, an event is triggered. An event is represented by (x, y, t, p), where (x, y) is the coordinate of the pixel that generates the event, t is the timestamp of the event, and p is the polarity of the event, which indicates whether the change of intensity is positive or negative (i.e. dark to bright or bright to dark). This sensing mechanism allows the event camera to have a high temporal resolution, since sensing intensity is fast. Independent pixel operation and the four-field event representation result in low latency recording and data transfer. Moreover, since the events are movement triggered, they are sparse and asynchronous, which allows the

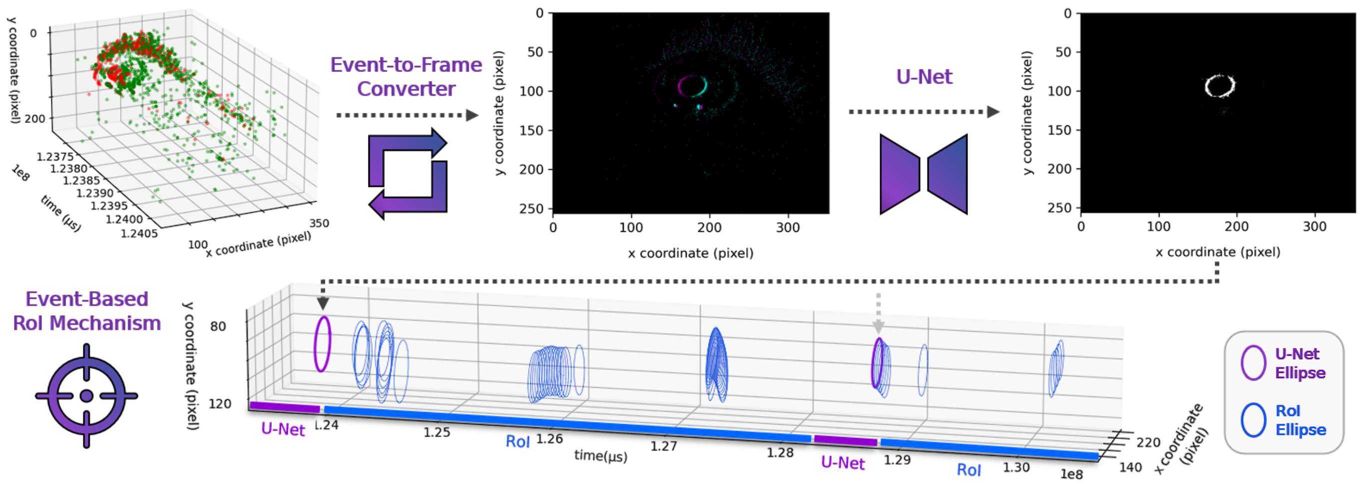

Fig. 1. Overview of the event-based eye tracking system. A stream of events recorded by near-eye event camera is the input, and the output is a stream of pupil ellipses. The event-to-frame converter encodes the eye motion expressed by the events into a 3-channel frame. A U-Net architecture classifies each pixel whether represents the pupil event or not. A pupil ellipse is fitted to the pixels and serves as the reference region for the RoI mechanism to accumulate pupil events and fit without inference. Another round of inference happens when the RoI mechanism looses track of the pupil. The purple ellipses are fitted from the pixels classified by U-Net, and the blue ones are fitted from the events accumulated with the RoI mechanism.

event camera to operate with low power. These characteristics make the event camera more favorable over RGB cameras in a high-speed eye tracking system with a low power budget.

B. Near-eye Event-Based Dataset

The dataset [10] used in this work was collected with DAVIS-346 camera from iniVation. It is a system that has an event camera and an RGB camera that can operate simultaneously with a resolution of 346×260 each. The camera is placed off-axis near the subject's eye with only the eye in the FoV. The dataset includes the recording of 24 subjects with a variety of eye characteristics and movements, including blinking, random saccades, and smooth pursuit. There are 330 seconds of event-based and frame-based data recordings per subject. Since the dataset is collected for gaze estimation applications, the labels in the dataset are the gaze points on the screen, instead of segmentation maps of the eye for eye tracking applications. Thus, we processed the frame-based data with the frame-based eye segmentation algorithm in [6] to get the ellipses representing pupil, which are compared against in the experimental result section. The events that are spatiotemporally close to the pupil ellipses in the frames are labeled as pupil events and serve as the ground truth for the model training.

III. Proposed Method

The proposed eye tracking system tracks the pupil location from the event-based data stream as shown in Fig. 1. The three functional blocks are elaborated on in this section.

A. Event-to-Frame Converter

The first step of the conversion is accumulating enough events that can represent eye movement. Instead of collecting events in a fixed time window, we collect a fixed amount

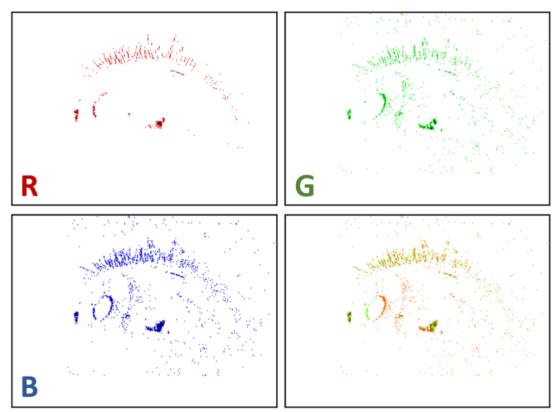

Fig. 2. The separated R, G, B channels and the combined 3-channel frame converted from an event set with 2000 events. The red channel has the positive polarity events and the green channel has the negative polarity events. The blue channel has all the 2000 events.

of events as a set. With the fixed time window method, the number of events in a set depends on eye movement. Thus, sets containing a small number of events fail to express the eye features, while those with a large number cause the eye features to overlap. On the contrary, the asynchronous fixed number of events method that we employ ensures all the event sets represent the same amount of motion with clear eye features. We empirically find that accumulating 2000 events in a set results in good system performance.

The second step is to convert a set into a frame as shown in Fig. 2. Each eye component has its distinctive spatiotemporal distribution. Thus, encoding the distribution into a 3-channel frame is crucial. The eyelash has a loose, fog-like event distribution in both polarities, and the events representing the eyelid are dense within and across polarities. The pupil

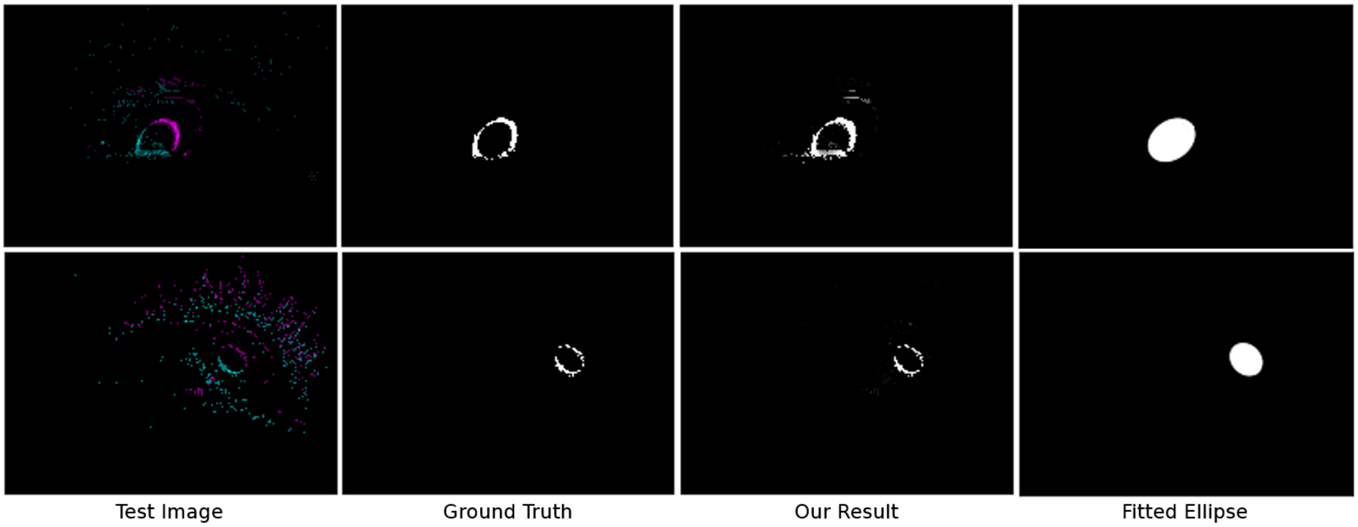

Fig. 3. Examples of pupil event segmentation results by the U-Net architecture and the fitted ellipse. The model is able to successfully extract pupil events from different sizes of pupils at different locations with different event densities, and the pixels can be correctly fitted to ellipses that represent the pupil.

events are dense within each polarity, but the negative and positive event clusters are wide apart. The distribution of the events within a polarity and the relationship between the two polarities are important to describe eye parts. Therefore, we generate a 3-channel frame with the first channel for positive events in the set, the second channel for negative events, and the third channel for all the events combined. Each event contributes an intensity value of 10 for its pixel coordinate in the polarity channel it belongs to and the third channel. The maximum intensity of each pixel is 255, and no normalization is performed, since the value represents the number of events that are triggered at the pixel coordinate.

B. Pupil Event U-net

The CNN that extracts the pupil events from the converted frame is U-Net [13], [14]: an encoder-decoder architecture with skip connections for semantic segmentation. There are 17 layers and 466,562 parameters in the network. The input layer has a size of $352\times256\times3$, and the frames converted from events are center padded and clipped to match the size. In the encoding path, there are two convolutions with ReLU and max pooling pairs and one convolution with relu at the end. The decoding path consists of two sets of up-convolutions, concatenate and convolution with relu followed by convolution with softmax activation. The concatenation stacks the intermediate results of the encoding path with the decoding layers. The output has 2 channels with the same size as input 352×256. In the first channel, the pixels that are set to 1 represents the location of the pupil events, and in the other channel, the pixels set to 1 are the background. Compared to the number of background pixels, the number of pupil pixels is significantly less. Thus, the model can easily achieve a small loss by predicting the entire image as the background. To overcome this problem, we incentivize the model to predict the pupil pixels correctly by using a weighted loss of each output channel. The loss of the output channel with pupil pixels is given a weight of 0.9, and the other channel is set to 0.1. The predicted output pixels with a value above 0.9 are classified as pupil pixels and are fit to an ellipse [15] representing the pupil feature.

C. Event-based Region-of-Interest (RoI) mechanism

Applying the CNN inference on each event set is the latency and power bottleneck of the system. To optimize the performance, we propose an RoI mechanism to track the pupil without performing inference on each collected event set. Once the pupil location is identified with U-net from the first event set, we mark a region of interest where the incoming pupil events are likely to land. We widen the ellipse by 8 pixels and mark the toroidal band as the RoI. Instead of accumulating a set of 2000 events that are triggered at any coordinate in the FoV, we monitor only the RoI and collect 256 events that are triggered within this region. The events in the RoI set are then fit into an ellipse that updates the pupil location for the next iteration. This region-based tracking mechanism leaves two situations in which a CNN inference would be required. The first is when the system initially begins its first iteration, and the second is when an eye blink results in the pupil being invisible. Thus, in an event data stream, the total number of CNN inferences required is drastically reduced by leveraging the high temporal resolution and movement-triggered characteristic of the event camera to reliably keep track of the pupil. In contrast, using the same mechanism with an RGB camera is infeasible, since it would lose track of the pupil when the eye moves fast between successive frames.

IV. Experimental Results

A. Pupil Feature Extraction

Fig. 3 shows two examples of CNN-based pupil segmentation. We trained the U-net on the event-based data of 24

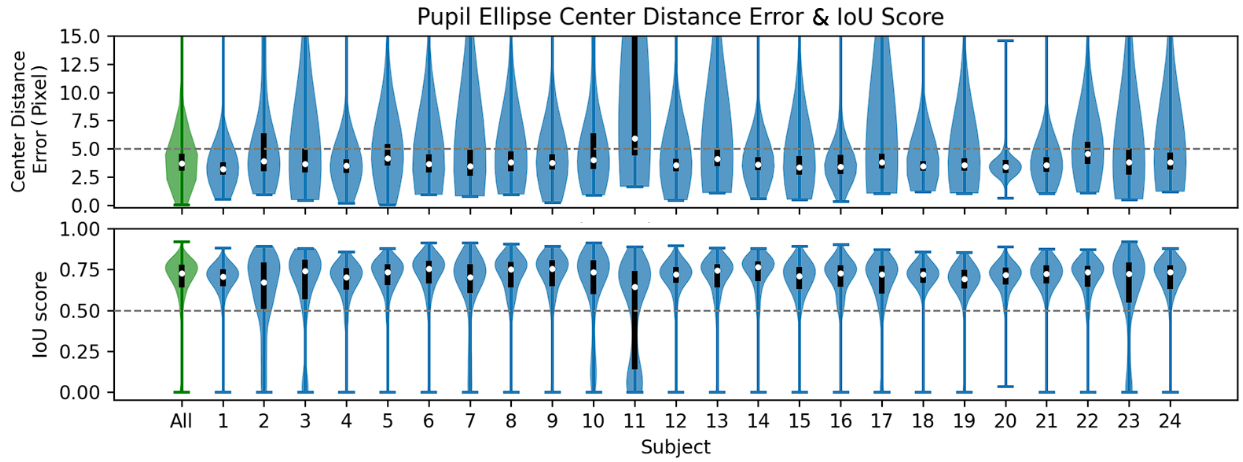

Fig. 4. The violin plot and box plot of the pupil ellipse center distance error (top) and IoU score (bottom). The ellipse fitted from the pixels classified as pupil events by U-Net is compared against the ground truth ellipse from the frame. Most of the subjects have a 3rd quartile pixel distance error that is under 5 pixels with the median around 3 pixels. The IoU scores of the majority of the subjects are above the target good score 0.5, and the medians are above 0.7. The first column is the aggregation of the result from all the subjects, and the overall median center distance error is 3.6 pixels and the median IoU score is 0.72. The plots show the effectiveness of the event-to-frame converter and the pupil event U-Net across subjects.

subjects that is split 70%-15-%-15% into training, validation and testing sets. We use weighted categorical cross-entropy loss function and Adam optimizer that has exponential decay with an initial learning rate of 0.001, a decay step of 4500, and a decay rate of 0.5. Two metrics are used to evaluate the eye tracking performance of the CNN: IoU (intersection over union) and center distance error, as shown in Fig. 4. IoU score is the overlap of the ground truth and the prediction ellipses over the union of the two masks. The overall IoU score has a median of 0.72 and is well above a good score (> 0.5). The center distance error of which the median is 3.68 pixels is the difference of the center of the ground truth and the prediction.

B. RoI Mechanism

On average, there are 7,130 frames and 20,669,767 events in a 330 second long data stream. Thus, a frame-based eye-tracking system has to perform 7,130 inferences. An event-based system that collects 2,000 events as a set has to run inference 10,334 times, which is more than the corresponding frame-based system. However, with the RoI mechanism, the inference count is reduced by 96.1% to 404 inferences and tracks the eye with only ellipse fitting on more than 99% of the pupil ellipses. Overall, our event camera based system requires 17.65$\times$ lesser inferences compared to the frame-based system.

C. System Performance

We evaluate the event-based eye-tracking system performance on multiple hardware platforms (Table I). The inference latency of the trained U-Net model (*TensorFlow or TF-Lite format*) is measured on each hardware platform. The energy consumption is estimated by assuming device operation at peak power while performing inference. For the system-level energy calculation, we consider the hardware to be active during each inference and idle during the lightweight RoI-based tracking between two inferences.

We observe that the energy consumption of CPU/GPU platforms hinders their usage for eye-tracking applications. This necessitates the use of specialized hardware like the edge-TPU for energy efficiency and low latency. We demonstrate the feasibility of our algorithm on the Google Coral Dev Board comprised of a CPU with an integrated ML accelerator. We achieve a U-Net inference latency of *66.4 ms* and estimate the overall eye-tracking system power of *160 mW* on this edge platform, thus demonstrating the feasibility of our algorithm.

TABLE I
E-Track Performance for U-Net Inference and Overall System with RoI Mechanism

Model Format / Hardware	**Inference**			**System**	
	Latency (ms)	*Power (W)*	*Energy (J)*	*Power (W)*	*Energy (J)*
TensorFlow / CPU (Ryzen9 5900H)	69.5	35	2.43	2.97	982
TensorFlow / GPU (RTX 3070)	4.1	125	0.51	0.62	206
TF-Lite / Coral (USB)	95.2	2	0.19	0.23	76.9
TF-Lite / Coral (Dev Board)	66.4	2	0.13	0.16	53.6

V. Conclusion

This work presents the first completely event camera based eye tracking system. Operating without additional RGB camera or IR LED reduces the hardware footprint and power consumption. The event-to-frame converter and U-Net CNN accurately locate the pupil from the event data stream, while the RoI mechanism effectively reduces the system workload. We demonstrate feasibility by deploying the system on an edge accelerator SoC. Our proposed event-based eye-tracking system enables low-latency asynchronous eye tracking on the power-limited and latency-sensitive XR headsets.

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